University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

Professor: Hadi Salmasian

Final Exam

December 7, 2013

Surname _____ First Name _____

Student # _____

Instructions:

(a) You have 3 hours to complete this exam.

(b) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.

(c) Write your student number at the top of each page in the space provided.

(d) No notes, books, scrap paper, calculators or other electronic devices are allowed.

(e) You are strongly recommended to write in **pen**, not pencil.

(f) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1–14	15	16	17	18	19	20	21	22	Total
Maximum	28	4	2	3	4	6	6	4	5	62
Grade										

Part A: Answer Only Questions

In Questions 1–14, only your final answer will be considered for marks. Each question is worth 2 points.

1. Let $A = \begin{bmatrix} -1 & 0 \\ -\frac{1}{2} & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & \frac{3}{2} \\ -1 & 0 \\ 2 & 1 \end{bmatrix}$. Compute AB^T.

Answer:_____

2. Let $z = 3i - 1$ and $w = 3 + i$. Write the complex number $\dfrac{z + \overline{w}}{w}$ in the form $a + bi$ where a and b are real numbers.

Answer:_____

3. Let $A = \begin{bmatrix} 3 & -2 & -2 & 4 \\ 0 & -2 & 0 & -\frac{1}{2} \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & -2 \end{bmatrix}$. Write down $\operatorname{rank}(A)$ and the dimension of $\operatorname{Nul} A$.

Answer:_____

4. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 3 & 0 \\ 0 & 1 & -\frac{1}{2} \end{bmatrix}$. Write down the eigenvalues of A^3 and A^{-1}.

Answer:_____

5. Let A be an invertible $n \times n$ matrix. Which of the following statements are false?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) The matrix equation $A\mathbf{x} = \mathbf{0}$ is consistent.
(b) The scalar $\lambda = 0$ is an eigenvalue of A.
(c) $\text{rank}(A) = n$.
(d) The matrix A^2A^T is invertible.
(e) A is necessarily diagonalizable.

Answer:_____

6. Let A, B, and C be 3×3 matrices such that $\det(A) = -\frac{1}{2}$, $\det(B) = \frac{1}{3}$, and $\det(C) = 4$. Calculate $\det(-A^TB^{-1}A^2CA^{-1})$.

Answer:_____

7. Let $A = \begin{bmatrix} i & i+1 & \frac{i}{i+2} & 3 \\ 0 & -\frac{1}{i} & \frac{1}{2}+i & 2 \\ 0 & 0 & -i^2 & -2i+3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$. Write down the eigenvalues of A and their multiplicities.

Answer:_____

8. Let A be an $m \times n$ matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^m$. Which of the following statements implies that the matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) Every row of A contains a pivot position.
(b) $\operatorname{rank} A = n$.
(c) $m = n$ and A is invertible.
(d) $m < n$.
(e) $m = n$.

Answer:_____

9. For each of the following subsets of $\mathbb{R}^3$, determine if it is a subspace or not.
If the given set is a subspace, you should write **yes**. if it is not a subspace, you should write **no**.

(a) $H = \left\{ \begin{bmatrix} a+b \\ b+c \\ c-a \end{bmatrix} \;\middle|\; a, b, c \in \mathbb{R} \right\}$

(b) $H = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \;\middle|\; x, y, z \in \mathbb{R} \text{ and } x + y \geq 0 \right\}$

(c) $H = \left\{ \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ -1 \end{bmatrix} \right\}$

(d) $H = \left\{ \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \;\middle|\; x_1, x_2, x_3 \in \mathbb{R} \text{ and } x_1 = x_2 + x_3 \right\}$

Answer:_____

10. Write down the characteristic polynomial of the matrix $A = \begin{bmatrix} 1 & 1 & -5 & 4 \\ 0 & -2 & 3 & 3 \\ 0 & 0 & 2 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Answer:_____

11. Let A be a 10×8 matrix and $\mathbf{b}$ be a vector in $\mathbb{R}^{10}$ such that the equation $A\mathbf{x} = \mathbf{b}$ has a unique solution. Which of the following statements cannot be correct?

More than one answer may be correct. You should indicate **all** of the correct answers. You will receive negative credit for each incorrect answer.

(a) $\operatorname{rank} A = 10$.
(b) The columns of A are linearly independent.
(c) Every row of A contains a pivot position.
(d) $\dim \operatorname{Nul} A = 2$
(e) The equation $A\mathbf{x} = \mathbf{0}$ is consistent.

Answer:_____

12. Suppose that $\begin{vmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} = -2$. Calculate the following two determinants.

$\begin{vmatrix} a_{2,1} & a_{2,2} & a_{2,3} \\ a_{1,1} & a_{1,2} & a_{1,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{vmatrix} =$ **Answer:**_____

$\begin{vmatrix} a_{1,1} & 2a_{1,2} & a_{1,3} - 2a_{1,1} \\ a_{2,1} & 2a_{2,2} & a_{2,3} - 2a_{2,1} \\ a_{3,1} & 2a_{3,2} & a_{3,3} - 2a_{3,1} \end{vmatrix} =$ **Answer:**_____

13. Let $A = PBP^{-1}$ where $P = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. Compute A^5.

Answer:_____

14. Determine <u>all</u> values of the parameter p for which the following linear system is <u>inconsistent</u>.

$$\begin{cases} -2x_1 + 3x_2 - x_3 = 2 \\ x_2 + x_3 = -1 \\ -2x_2 + px_3 = -3 \end{cases}$$

Answer:_____

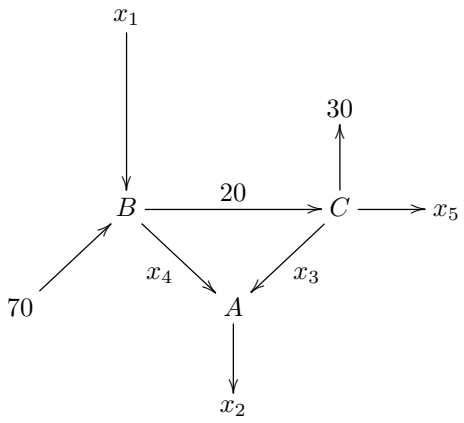
Part B: Long Answer Questions

In Questions 15–22, you must show your work and justify your answers to receive full marks. Partial marks may be awarded for making sufficient progress towards a solution.

15. **(4 points)** Is the following linear system consistent or inconsistent? If it is consistent then find its general solution and write it in vector parametric form.

$$\begin{cases} x_1 + 5x_2 + 2x_3 = x_4 \\ x_3 + 2 = 2x_4 \\ 3x_1 + 15x_2 + 4x_3 + x_4 = 4 \end{cases}$$

16. **(2 points)** The traffic flow in a city is represented by the diagram below. The arrows indicate the directions of one-way traffic. Numbers and variables indicate the number of cars per hour in the corresponding direction. Write down a linear system describing the traffic flow. Do not solve the linear system.



17. **(3 points)** Find all values of the parameter t such that the set of vectors

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 0 \\ -2 \end{bmatrix}, \begin{bmatrix} -3 \\ 2 \\ 4 \\ 3 \end{bmatrix}, \begin{bmatrix} 5 \\ 3 \\ 7 \\ -6 \end{bmatrix}, \begin{bmatrix} 1 \\ -4 \\ t \\ 1 \end{bmatrix} \right\}$$

is a basis for $\mathbb{R}^4$.

18. **(4 points)** Let $A = \begin{bmatrix} 1 & -2 & \frac{1}{2} \\ -2 & 4 & s-1 \\ 2 & -4 & 1 \end{bmatrix}$. First determine the value of s such that $\dim \operatorname{Nul} A = 2$. Then find a basis of $\operatorname{Nul} A$ for this value of s.

19. The city of Chicago will never forget the first week of November: every single day, the controversial statements by the city's mayor made it to the morning news headlines. Surveys indicate that, every day, 40% of the mayor's supporters would change their minds and become his opponents, whereas 20% of his opponents would change their minds and become his supporters.

(a) **(1 point)** Write the migration matrix M.

(b) **(2 points)** Surveys show that Monday morning, before the mayor's first controversial statement, exactly half of Chicago's population were the mayor's supporters, and the remaining half were his opponents. Determine the percentage of his supporters and opponents on Tuesday evening.

(c) **(3 points)** Assuming that the mayor did not stop making controversial statements on a daily basis, what would be the percentage of his supporters and opponents in the long run?

20. (a) **(2 points)** Let $A = \begin{bmatrix} 3 & 1 & -2 \\ 0 & -1 & 0 \\ -1 & 5 & 4 \end{bmatrix}$. Write the characteristic polynomial of A and compute its eigenvalues.

(b) **(4 points)** Let $B = \begin{bmatrix} 4 & -1 & 1 \\ 2 & 1 & 2 \\ -1 & 1 & 2 \end{bmatrix}$. Find a basis for the eigenspace of B that corresponds to the eigenvalue $\lambda = 3$.

21. **(4 points)** Compute the inverse of $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 3 \\ \frac{2}{3} & 1 & 2 \end{bmatrix}$.

22. An economy has two sectors: Agriculture and Energy. We know that:

- In order to produce one unit of output, the Agriculture sector should consume .3 units from itself and .4 units from the Energy sector.
- In order to produce one unit of output, the Energy sector should consume .7 units from itself and .5 units from the Agriculture sector.

(a) **(1 point)** Write down the consumption matrix for this economy.

(b) **(1 point)** Determine the intermediate demands that are created if Agriculture wants to produce 15 units and Energy wants to produce 20 units.

(c) **(3 points)** Calculate the production levels that are required to meet a final demand of 5 units from the Agriculture sector and 2 units from the Energy sector.

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